

Family-Conditioned Multiview Fitting for Semantic Primitive Proxy Actors in UAV Digital Twins (SPPA-MVFit)

Authors omitted for review

Abstract

Object-level UAV telemetry is not a mesh: a detector may report a class family, confidence, track identifier, bounding region, and approximate pose, but no occupancy geometry. Geospatial digital twins that maintain live object layers nevertheless need a per-object 3D representation that is cheap to update in place and explicit about uncertainty. This paper studies whether a frozen semantic-family part graph improves lightweight 3D occupancy relative to an input-matched, equal-budget nonsemantic graph when both consume the same calibrated top and side silhouettes. The method, family-conditioned SPPA-MVFit, aligns five shared parameters of a fixed eight-part primitive actor with a 31-candidate silhouette objective. On a preregistered developer-held-out synthetic test of 240 actors balanced across six morphology families and two source strata, the paired mean voxel-IoU difference over Generic-MVFit is 0.190 (95% CI [0.181, 0.199]), exceeding the prespecified +0.030 margin, and the margin holds under all prespecified mask and morphology corruptions. Median fitter wall time is 9.4 ms on the benchmark machine. A post-hoc sanity check on 52 external real meshes shows the margin does not transfer (SPPA 0.413, generic 0.370, capsule 0.492, visual hull 0.656); occupancy leadership outside the design distribution is not claimed. An adversarial stratum that violates the family priors while preserving semantic class destroys one third of the advantage (+0.209 to +0.141) with identifiable tie zones, and a real detector stream case study (1,902 detections, exact tower-anchor ground truth) shows the field operating point is observation-bound and trades 2D image overlap for 3D part structure. A descriptor/update contract and an optional Unreal Engine backend are reported as deployment context.

Keywords: semantic 3D proxy; multiview silhouette fitting; UAV digital twin; family-conditioned part graph; primitive actor; occupancy IoU; geovisualization of dynamic objects.

1 Introduction

Urban and geospatial digital twins increasingly maintain live object layers—vehicles, field equipment, temporary structures—alongside static city geometry [2, 39]. Each of those layers needs a per-object 3D representation that is cheap to update in place, explicit about uncertainty, and legible at operational zoom levels rather than photoreal. Semantic-telemetry UAV twins make this problem concrete: a UAV-side perception stack reports object tracks, classes, confidence, positions, and approximate geometry, and the ground station must render those objects in an Unreal/Cesium 3D Tiles twin. That architectural choice creates a rendering problem that is separate from the flight system itself: object-level telemetry is not a mesh.

Curated 3D assets are useful when the class is known, the asset library is complete, and run-time loading is acceptable. Those assumptions are weak for field UAV operation. Long-tail labels, ambiguous detections, partial views, occlusion, motion blur, and changing object proportions make a fixed asset library brittle. Neural image-to-3D and text-to-3D methods address a different objective: visual asset generation. They can produce richer geometry, but they depend on usable crops or prompts, GPU inference, and mesh outputs that may be too dense for a portable flight laptop that is also running Unreal, telemetry, video, and tracking under a shared rendering budget.

SPPA proposes a lower-fidelity representation point. Instead of reconstructing the real mesh, it compiles a detection into a semantic part graph made from primitive boxes, cylinders, ellipsoids, cones, and tori. A vehicle becomes a cab, body/cargo region, wheels, and optional top details. A quadruped becomes a torso, head, and legs. A tree becomes a trunk and canopy volumes. Unknown or unsupported labels are not hallucinated into specific objects; they fall back to a generic uncertainty-aware display volume with explicit uncertainty metadata.

The question studied here is therefore not whether SPPA produces prettier geometry than neural 3D generation, but a measurable spatial-proxy question: does a frozen semantic-family part graph improve lightweight 3D occupancy relative to an input-matched, equal-budget nonsemantic graph when both consume the same calibrated top and side silhouettes? A role-labeled primitive proxy turns detection-level semantics into displayable, analyzable 3D geometry with bounded cost; this paper isolates the occupancy value of the semantic part structure behind that representation under controlled synthetic conditions.

The contribution is **family-conditioned SPPA-MVFit**: a preregistered, equal-budget comparison showing that the family graph improves clean voxel IoU over a generic eight-part graph by 0.190 (95% CI [0.181, 0.199]) on 240 held-out synthetic actors, together with boundary analyses that map where the prior helps, where it hurts, and where it stops transferring. The optimizer is deliberately a shared five-parameter global alignment and is not claimed as a novel fitting algorithm; runtime persistence and updateability are reported as separate engineering properties rather than collapsed into a utility score. Concretely, the paper contributes: (i) the comparison protocol itself—an input-matched, equal-budget, preregistered design that isolates the knowledge representation as the single varied factor; (ii) a quantified boundary map of that representation—wrong-family tokens, adversarial family instances, single-view degradation, external real meshes, and a real detector stream—showing measurably where the prior stops helping; and (iii) the lightweight actor contract (role labels, bounded updates, telemetry-scale descriptor), whose operational value is measured by a part-query task rather than asserted.

SPPA: language-assisted part decomposition compiled into deterministic 3D primitives

Language can draft the recipe offline; runtime compiles a reviewed versioned recipe plus telemetry.

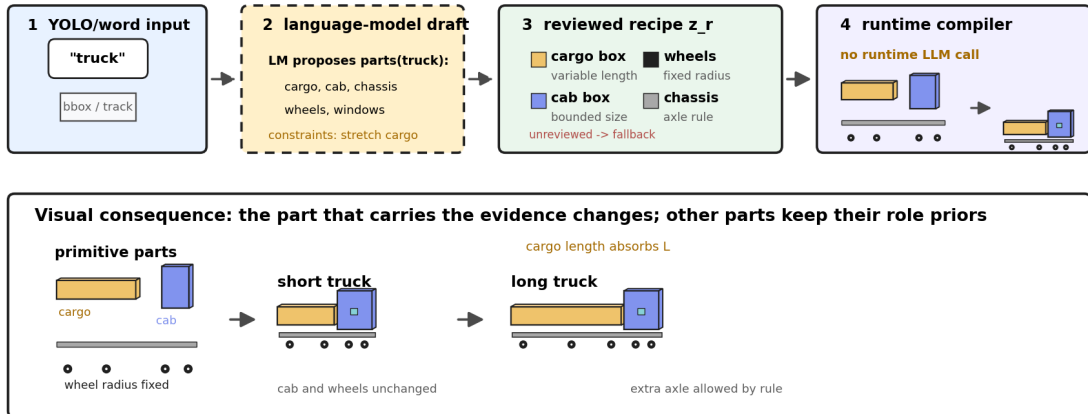


Figure 1: SPPA semantic-label-to-parts-to-3D mechanism for the truck archetype. A language model, ontology tool, or human author may draft candidate part roles offline, but runtime execution uses only a reviewed versioned recipe z_r . An input label activates an offline draft, the draft is reduced to reviewed part roles and bounded update rules, and runtime compiles those roles into primitive geometry. The short and long proxy variants illustrate the implemented role-specific rule: cargo length changes while cab and wheel priors remain bounded, so the object is not represented by root-scaling a single box.

2 Related Work

Text-to-3D and image-to-3D generators target visual asset fidelity from prompts or images [41, 29, 37, 23, 51, 27, 60, 47, 5, 48, 49, 54, 55, 56, 30, 26, 59, 58, 63, 50, 20]. SPPA targets a different runtime point: deterministic actor generation from sparse semantic telemetry, low polygon counts, and cheap per-frame updates.

Monocular and open-vocabulary 3D detection estimate 3D boxes or object state from images [6, 61]; YOLOE supplies approximate open-prompt detection/segmentation evidence rather than universal recognition [53]. Animal pose and reconstruction methods estimate detailed skeletons or category-specific 3D shape [62, 28, 18]. These methods can provide stronger evidence to SPPA, but SPPA itself is a rendering/backend layer, not a detector or pose estimator.

Category-conditioned parametric templates are the closest established paradigm. 3D morphable models fit a low-dimensional learned face prior to image evidence [4], and articulated body and animal models such as SMPL and SMAL fit a category-conditioned parametric template to scans or images [32, 65]. SPPA-MVFit instantiates the same pattern—a family-conditioned template aligned to observation evidence—with an eight-part primitive graph and a deliberately shared five-parameter global alignment. The operating point differs: those models target surface recovery of a specific category, whereas SPPA targets millisecond, track-updateable display actors from sparse UAV telemetry, where the contribution is the bounded runtime contract, the role labels, and the update semantics rather than surface fidelity.

Primitive representations have a long history, from volumetric primitives, recognition-by-components, and superquadrics to learned part decomposition and neural primitive/CSG parsing [33, 3, 46, 40, 52, 35, 13, 14, 34, 64, 45]. Procedural modeling, shape grammars, and learned shape programs generate or abstract structured objects from compact programs [36, 21, 22, 24, 43].

SPPA uses these ideas operationally, but the claimed novelty is the bounded, track-updateable semantic telemetry contract inside an Unreal UAV digital twin.

Recent scene-level systems narrow the gap from other directions. SAM 3D decomposes a single image into composable per-object geometry with layout [44], and PartCrafter generates part-structured meshes from one image [30]; both are heavy offline reconstruction or generation pipelines, not millisecond telemetry-driven proxies with an update contract. SuperQuadricOcc reaches real-time semantic occupancy with superquadric volume rendering in driving scenes [16], and lightweight urban twins render proxy building geometry at interactive rates [31]; neither constructs role-labeled, metric-scaled object actors from detector tags and sparse silhouettes. Test-time spatial control steers generative 3D models with coarse primitives [12], which complements rather than replaces a deterministic proxy contract.

Graphics systems already provide level-of-detail, impostors, billboards, streaming tiles, and virtualized geometry [7, 8, 11, 39]. 3D and dynamic scene graphs maintain semantic world structure [1, 42, 57], and Hydra-style systems build and optimize hierarchical scene graphs online for mapping and planning [19]. SPPA deliberately keeps its own lightweight contract instead of becoming a node type in such a substrate: a proxy actor is a display-side object with a bounded descriptor/update payload, not a mapping layer. Occupancy grids, OctoMap, ESDF, and TSDF maps support planning and mapping [17, 38]. SPPA should be read as a display actor representation that can consume such upstream evidence, not as a mapping substrate.

3 Materials and Methods

3.1 The SPPA contract and part graph

At time t , SPPA consumes an object state

$$x_t = \{i, \ell, c, b_t, m_t, \xi_t, s_t, \Sigma_t, h_t\},$$

where i is a track identifier, ℓ a detector label, c confidence, b_t a bounding region, m_t an optional mask or footprint, ξ_t a world pose estimate, s_t optional metric dimensions, Σ_t uncertainty, and

h_t the track history. The resolver maps ℓ to one of three cases: an `exact_class` label with a fixed archetype; a `keyword_archetype` mapping of an out-of-template label to a finite reviewed archetype, such as vehicle or vertical structure; or a `fallback_unknown_label` rendered as a generic uncertainty-aware volume. The versioned recipe manifest covers 23 archetypes and 95 exact or keyword labels; SPPA can accept an open detector tag at runtime, but the tag is normalized into a reviewed family or a conservative unknown fallback before geometry is emitted.

Each archetype a defines a part graph

$$G_a = (V_a, E_a), \quad v_k = (p_k, r_k, \tau_k, \rho_k),$$

where p_k is a primitive type, r_k is its role, τ_k its transform relative to the actor root, and ρ_k role-specific parameters such as dimension priors, color/material role, and confidence tags. The current implementation uses boxes, cylinders, ellipsoids, cones, and tori (Figure 2).

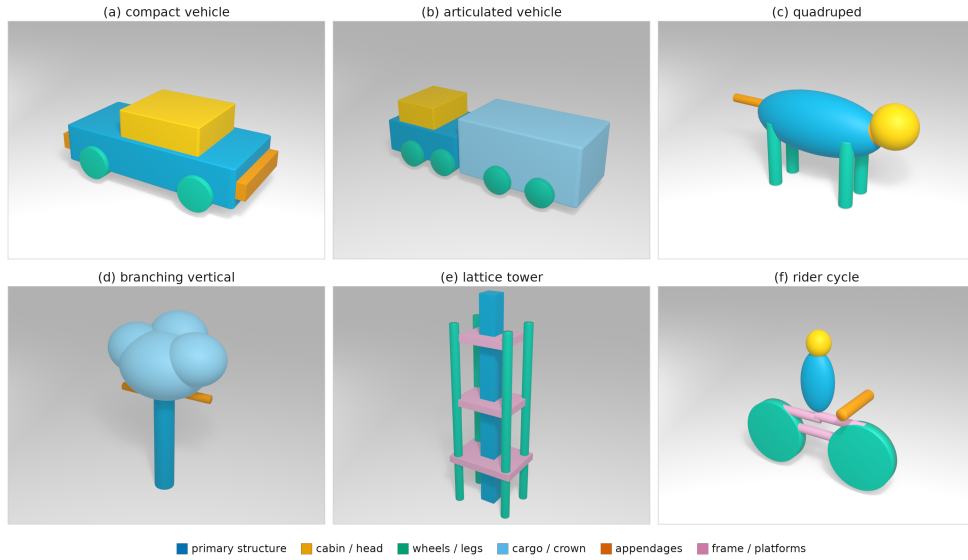


Figure 2: The six benchmark family part graphs used by SPPA-MVFit (`compact_vehicle`, `articulated_vehicle`, `quadruped`, `branching_vertical`, `lattice_tower`, `rider_cycle`): eight primitive slots per family with role-conditioned priors; colors denote semantic roles.

Vehicle adaptation is the clearest implemented case. A long truck and a short truck should not be produced by globally scaling the same box. In the current vehicle grammar, cab and wheel dimensions remain bounded by role priors, while the cargo/body span and axle placement absorb length changes. In the synthetic truck check, two descriptors with equal width/height and different length keep cab scale fixed at $[1.656, 2.070, 1.952]$ m and wheel scale fixed at $[0.392, 0.094]$ m, while cargo/body length changes from 2.263 m to 5.188 m.

Language model use (offline only). A language model, ontology tool, or human author may propose candidate part decompositions offline; those candidates must be reviewed, versioned, and regression tested before admission. No language model is called at runtime.

Pose, yaw, and updates. The descriptor separates actor pose ξ_t from part parameters q_t . Pose changes are per-frame updates; topology regeneration is avoided unless the class/archetype changes or new evidence crosses a configured threshold. Yaw ambiguity is handled explicitly: a top-down footprint or PCA orientation gives an axial orientation modulo π , and signed yaw requires velocity, asymmetry, multi-frame evidence, or an external heading source; otherwise the descriptor sets `yaw_ambiguous=true`.

3.2 End-to-end pipeline: two input paths, one builder

Both input contracts terminate in the same deterministic builder; they differ only in how much evidence reaches the recipe. The tag/text path compiles a reviewed recipe at its prior dimensions: normalize the label through the reviewed ontology, select the frozen part graph G_a and its prior role parameters, emit role-labeled primitives with connector cylinders for semantically attached parts, and emit the versioned descriptor consumed once per track by the renderer. The detector/image path inserts four evidence stages before the same builder: (1) **Observe**—detector label, confidence, bbox, and native mask polygon are stored in a `SPPA-OBS-0.1` record with declared camera/flight geometry and a covariance-style uncertainty envelope; (2) **Project**—the mask polygon is projected through the declared camera model into an oriented ground footprint, keeping the axis-aligned mask box and a PCA axial orientation as audit evidence; (3) **Scale**—with declared metric calibration the footprint becomes a metric length/width/height proposal; without calibration the mask remains image-space evidence and cannot change metric geometry; (4) **Gate**—a class-aware constraint-fusion step accepts plausible scale, clamps implausible length/width ratios through the family aspect prior, withholds unsafe image-derived pose/yaw from the runtime descriptor, and records every decision with its reason. Because both paths share the builder and the descriptor contract, their outputs are directly comparable (Figure 3).

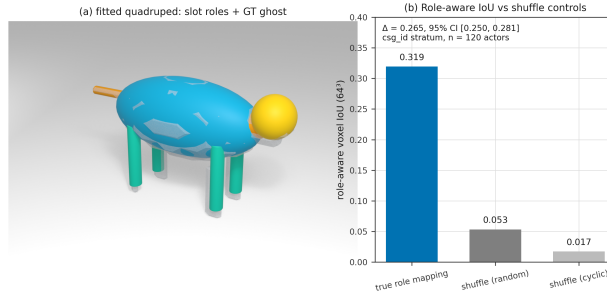


Figure 3: (a) A built SPPA proxy with primitives colored by semantic role. Role labels are part of the emitted actor contract: they drive bounded per-role update rules, material/uncertainty tags, and the connector primitives between semantically attached parts. (b) Role-aware IoU of the fitted proxies against random-shuffle and cyclic-shuffle controls (0.319 versus 0.053 and 0.017; Section 4.5).

Worked example (tractor-trailer probe). The real crop yields YOLOE evidence `vehicle` at confidence 0.47 with a 54-point native mask. Projection proposes a raw footprint of $16.85 \times 6.21 \times 3.40$ m; the gate applies soft low-confidence fusion, clamps the footprint through the vehicle aspect prior to $12.11 \times 3.76 \times 3.40$ m, withholds image-derived pose, and refines the weak label only to the conservative `articulated_vehicle` family. The shared builder then emits a 1,904-triangle proxy (Supplementary Section S.4).

3.3 Descriptor and runtime backend

A `SPPA-DESC-0.2` record stores resolver source, match type, fallback reason, scale source, mask footprint evidence, yaw ambiguity, shape confidence, material source, part list, primitive counts, and scheduler metadata; a `SPPA-UPD-0.2` record separates create/regenerate payloads from pose/no-op/shape updates, and `descriptor_id` acts as a topology-cache identity so that a pose update never looks like a new mesh descriptor. The design goal is to generate once per track/archetype, update pose/material/uncertainty every frame, and apply shape-parameter changes in place when the part topology is unchanged. An optional Unreal Engine 5.7 backend ingests the same descriptors alongside curated assets, which remain the default; the SPPA backend constructs primitive components from the `parts` array, applies update packets, and rejects invalid JSON or mismatched descriptor identifiers without changing the current actor.

3.4 Confirmatory protocol

Every compared method receives the same case identifier, morphology-family token, metric world box, and two 96×96 orthographic binary silhouettes (top and side). No method receives RGB pixels, detector scores, source parameters, ground-truth occupancy, evaluation projections, or real or replay telemetry. An actor is an ordered list of eight primitive slots (boxes, cylinders, ellipsoids). Family graphs and a single **generic** graph are stored as static JSON. The only builder used by text-only SPPA, Generic-MVFit, and SPPA-MVFit is

$$A = \text{Build}(g, \theta), \quad \theta = (\log s_x, \log s_y, \log s_z, s_{\text{sec}}, o_{\text{sec}}), \quad (1)$$

with identical bounds for every graph g . Text-only SPPA uses the family graph at the default θ_0 . Generic-MVFit and SPPA-MVFit optimize the same five parameters with the same silhouette disagreement objective—the mean top/side disagreement $0.5(1 - \text{IoU}_{\text{top}}) + 0.5(1 - \text{IoU}_{\text{side}})$ plus a small centering regularizer—using a coordinate descent over three shrinking step fractions that evaluates exactly 31 objective values; only the graph identity differs (**generic** vs family). Baselines include axis-aligned box, ellipsoid, capsule, billboard, and nonsemantic visual hull from the same masks. The fitter is deliberately a five-parameter global alignment shared by all slots; the contribution is the family-conditioned representation and the runtime contract, not the optimizer (Figure 4).

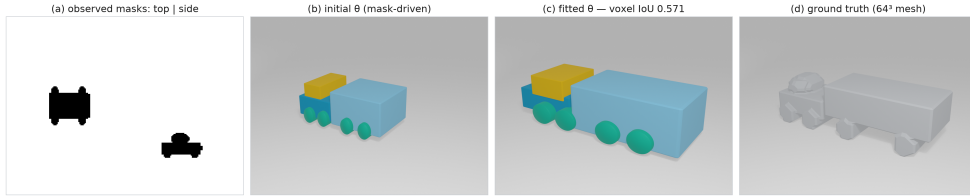


Figure 4: Fitting of SPPA-MVFit on a held-out actor: (a) the observed silhouette pair, (b) initialization from the observed mask extents, (c) the converged eight-part proxy, and (d) the ground-truth voxel mesh.

The **generic** graph was designed by the authors as a plausible family-agnostic prior: eight symmetric ellipsoid slots with no family-specific topology, proportions, or role labels, frozen before the benchmark was run. It is hand-crafted, not learned, not random, and not the mean of the six family graphs. Because the same team authored the family graphs and this generic competitor, three alternative generic graphs were registered in writing before any measurement and are evaluated in Section 4.6; the shared-authorship threat is discussed in Section 5.

The primary hypothesis compares clean-mask 64-cubed voxel IoU of SPPA-MVFit against Generic-MVFit. Analysis units are 240 source actors (20 per family in each of two strata). The confirmatory estimand is the equal-stratum mean paired difference. H1 passes only if the lower endpoint of a two-sided 95% stratified actor-bootstrap percentile interval (10,000 resamples, seed 77157) exceeds +0.030. The margin and the power analysis behind the sample size were fixed before evaluation: with paired-difference SD 0.12, $n=240$ actors give approximately 90% normal-approximation power to distinguish a true mean of +0.055 from the +0.030 margin. The full protocol was preregistered and sealed before evaluation; the seed provenance, amendment log, and freeze records are given in the supplement. Development results were used only to implement the frozen package and are not confirmatory.

Source actors come from two strata: CSG-ID unions of solids with family topology, and implicit-OOD occupancy from superelliptic extrusions, taper/twist, toroidal sections, and spline-distance tubes. Source and method modules do not import each other. The held-out set is developer-held-out synthetic geometry, not an externally blinded public UAV benchmark.

All confirmatory and post-hoc fits ran on one benchmark workstation (AMD Zen 5, 32 logical CPUs, Windows 11, CPython 3.12.6); the external neural-generator runs of Section 4.8 used an RTX 5090 32 GB GPU on the same machine.

3.5 Evaluation data

The synthetic benchmark covers six families selected before freezing for morphological coverage of the ground-object envelope a UAV twin must display: compact rigid bodies (`compact_vehicle`), elongated articulated bodies (`articulated_vehicle`), legged organic forms (`quadruped`), branching vertical growth (`branching_vertical`), thin lattice structures (`lattice_tower`), and rider-plus-cycle combinations (`rider_cycle`), with 20 actors per family in each of the two strata ($n=240$). A post-hoc exploratory drop-one-family re-analysis (Section 4.3) confirms the headline difference is not carried by any single family: excluding any one family leaves the paired difference between 0.152 and 0.214, always above the +0.030 margin. As an external case study outside the design distribution, the same methods were run on $n=52$ real third-party meshes (Objaverse LVIS subset and ModelNet40) under a class-to-family mapping frozen in writing before any evaluation (Section 4.7). Four real-image probes (a road cyclist, a mountain electrical tower, a mountain tractor, and a tractor with trailer) exercise the dual-input contract of Section 3.2 with a YOLOE-26s open-vocabulary detector (checkpoint `yoloe-26s-seg.pt`) as approximate semantic evidence (Section 4.9).

4 Results

4.1 Primary confirmatory result

Table 1 reports the confirmatory decision: H1 passes, with mean $\Delta\text{IoU} = 0.190$ and 95% CI [0.181, 0.199]. The CSG-ID stratum mean difference is 0.209 and the implicit-OOD stratum mean difference is 0.172; neither is non-positive, so a cross-stratum generalization warning is not triggered by the protocol rule. Figure 5 shows the paired difference in every family-by-stratum cell (per-cell values in Supplementary Section S.9); all twelve cells are positive, and the smallest gain is the compact-vehicle implicit-OOD cell.

Table 1: Preregistered primary endpoint on the held-out synthetic test ($n=240$ actors). Provenance: `synthetic_geometry`. H1 passes only if the stratified bootstrap 95% lower bound exceeds +0.030.

Quantity	Value
Primary mean Δ IoU (SPPA-MVFit – Generic-MVFit)	0.190
Stratified bootstrap 95% CI	[0.181, 0.199]
Prespecified superiority margin	+0.030
H1 decision (lower bound > margin)	PASS
Actors (analysis units)	240
CSG-ID stratum mean Δ	0.209
Implicit-OOD stratum mean Δ	0.172
Resolution sensitivity 48/64/80	PASS

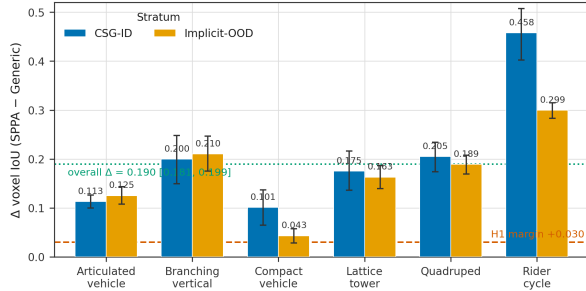


Figure 5: Paired clean-mask mean IoU difference of SPPA-MVFit minus Generic-MVFit by morphology family and source stratum ($n=240$). All twelve cells are positive; the equal-stratum aggregate is the primary estimate of Table 1.

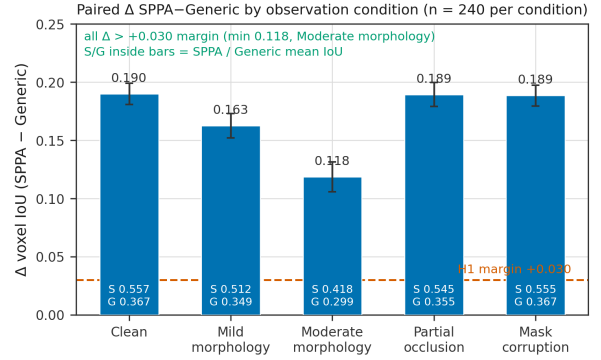


Figure 6: Paired SPPA-MVFit minus Generic-MVFit mean IoU difference with 95% bootstrap intervals under the five prespecified observation conditions ($n=240$ per condition). The dashed line marks the prespecified $+0.030$ margin; it is exceeded under every condition.

Table 2 reports clean-condition method means and wall-clock costs, and Table 3 the paired differences of SPPA-MVFit against the remaining baselines under the same stratified bootstrap construction. SPPA-MVFit exceeds SPPA text-only (mean $\Delta = 0.130$, CI $[0.116, 0.144]$) and all lightweight geometric baselines. Mean IoU of SPPA-MVFit (0.557) is slightly above nonsemantic visual hull [25] (0.522), but visual hull is denser occupancy geometry without a role-labeled lightweight actor contract, so it is not presented as a defeated runtime competitor. Median single-call descriptive SPPA-MVFit time is 9.4 ms (p95 10.6 ms), below the secondary local-cost gates of 100 ms median / 250 ms p95. A resolution-sensitivity check at grids 48/64/80 on the prespecified actor subset reports paired differences of 0.198, 0.197, and 0.188, with maximum deviation $|\Delta| \leq 0.0094$, below the 0.015 protocol threshold. The secondary intervals in Table 3 are per-comparison intervals and are not adjusted for multiplicity; confirmatory inference in this paper is H1 only.

Table 2: Clean-mask method summary on the held-out test ($n=240$). Mean/median are 64-cubed voxel IoU. Timings are single-call wall-clock milliseconds on the benchmark machine and are not Unreal frame times.

Method	Mean IoU	Median IoU	Median ms	p95 ms
SPPA-MVFit	0.557	0.563	9.43	10.60
Generic-MVFit	0.367	0.390	11.34	11.81
SPPA text-only	0.427	0.441	0.77	1.81
Visual hull	0.522	0.567	0.22	0.24
Ellipsoid	0.342	0.292	0.50	0.55
Capsule	0.325	0.260	0.95	1.07
Axis-aligned box	0.248	0.190	0.23	0.27
Billboard	0.173	0.148	0.23	0.26

Table 3: Paired clean-mask mean IoU differences of SPPA-MVFit minus each comparator (stratified actor bootstrap 95% intervals, per-comparison and unadjusted across secondary endpoints). Visual hull is a high-complexity geometry reference, not a lightweight runtime actor.

Comparator	Mean Δ IoU	95% CI low	95% CI high
Generic-MVFit (H1)	0.190	0.181	0.199
SPPA text-only	0.130	0.116	0.144
Visual hull	0.036	0.027	0.044
Ellipsoid	0.216	0.207	0.224
Capsule	0.233	0.224	0.241
Axis-aligned box	0.310	0.301	0.318
Billboard	0.384	0.375	0.393

4.2 Robustness under corrupted observations and failure analysis

The prespecified observation conditions—mild and moderate mask noise, partial occlusion, and moderate morphology corruption—were fixed in the protocol and evaluated with the confirmatory run (Figure 6; per-condition table in Supplementary Section S.9). The margin is exceeded under all five conditions, including moderate morphology corruption ($\Delta = 0.118$, CI [0.106, 0.131]). Moderate morphological corruption degrades the margin far more than mask corruption or partial occlusion ($\Delta = 0.189$ each): evidence that contradicts the graph prior hurts the fit more than missing or noisy evidence.

Exactly 5 of 240 clean cases fall below the prespecified failure threshold (voxel IoU < 0.25); all five are `lattice_tower` CSG-ID cases (worst 0.147 and 0.148). Inspection attributes the failures to sub-voxel geometry: the failing towers have leg and plate thicknesses of 0.09–0.37 world units against a $0.15 \times 0.10 \times 0.10$ voxel cell, and the seven actor methods score at most 0.24 on these actors (visual hull at most 0.43)—a resolution effect shared by every method, not a family-prior failure. Convergence telemetry supports the same reading: 88.7% of SPPA-MVFit cases still improve in the final sweep and its θ never sits at the frozen bounds, whereas Generic-MVFit pins θ at a bound in 57.9% of cases—evidence of a mismatched prior demanding out-of-range excursions, not of SPPA overfitting. Surface metrics agree with the volume ranking: Chamfer distance 0.008 for SPPA-MVFit versus 0.009 for visual hull and 0.016 for Generic-MVFit (Supplementary Section S.9), and F-score at 1.5 voxels of 0.831 versus 0.799 and 0.560 respectively (Table 4).

Table 4: Surface metrics on the clean condition ($n=240$). F-score at a 1.5-voxel threshold against ground-truth surfaces; paired differences versus SPPA-MVFit with stratified bootstrap 95% intervals.

Method	F-score@1.5 vox	Δ vs SPPA	95% CI
SPPA-MVFit	0.831	—	—
Generic-MVFit	0.560	−0.271	[−0.280, −0.262]
Visual hull	0.799	−0.032	[−0.040, −0.023]
SPPA text-only	0.706	−0.125	[−0.140, −0.110]

4.3 Representation versus fitting: the graph-by-fitting decomposition

A post-hoc exploratory analysis completes the graph-by-fitting factorial (Figure 7): the previously missing cell, the generic graph at the default θ_0 , scores 0.180. The family graph with no fitting at all (0.427) already exceeds the fully fitted generic competitor (0.367). Fitting adds +0.187 on the generic graph and +0.130 (CI [0.117, 0.143]) on the family graph, with a negative interaction (−0.058). The H1 headline therefore decomposes into a larger representation effect—graph effect at no fit +0.248, CI [0.234, 0.261]—plus a smaller fitting contribution on top of it, and the claim of this paper is accordingly about the family-conditioned representation, not about fitter sophistication. The drop-one-family re-analysis of Section 3.5 and the per-stratum effect estimates are tabulated in Supplementary Section S.9.

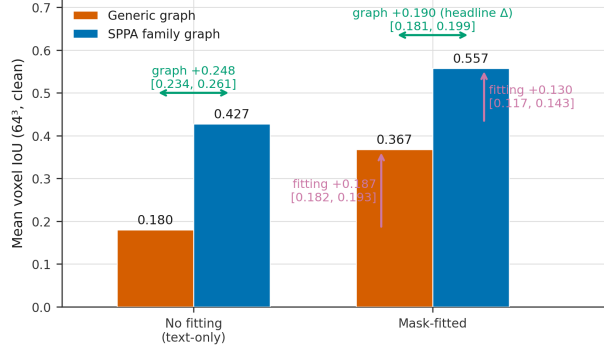


Figure 7: Graph-by-fitting 2×2 decomposition (clean condition, $n=240$, mean voxel IoU). Rows: graph (**generic** vs family); columns: no fitting (θ_0) vs 31-candidate silhouette fitting. The family graph without fitting already exceeds the fitted generic graph.

4.4 Wrong-family tokens

The confirmatory benchmark supplies the correct family token, so it does not measure the principal failure mode of a family-conditioned method. Each held-out actor was therefore also fitted under each of the five incorrect family tokens (1,200 additional deterministic fits through the same fitting path; Figure 8). A wrong token is far worse than no family prior at all: the wrong-token mean is 0.205 against Generic-MVFit 0.367, a paired difference of -0.162 (CI $[-0.167, -0.158]$), with 98.3% of actors scoring below their own generic fit, and even the per-actor *best* wrong token (0.361) does not beat the generic mean. The deployed contract already routes unsupported labels to a generic fallback volume and refines low-confidence detector labels only to conservative families; this result makes the symmetric design implication explicit—a family token should be accepted only with adequate detector confidence or agreement, and otherwise the fit should degrade to the generic graph rather than commit to a mismatched family.

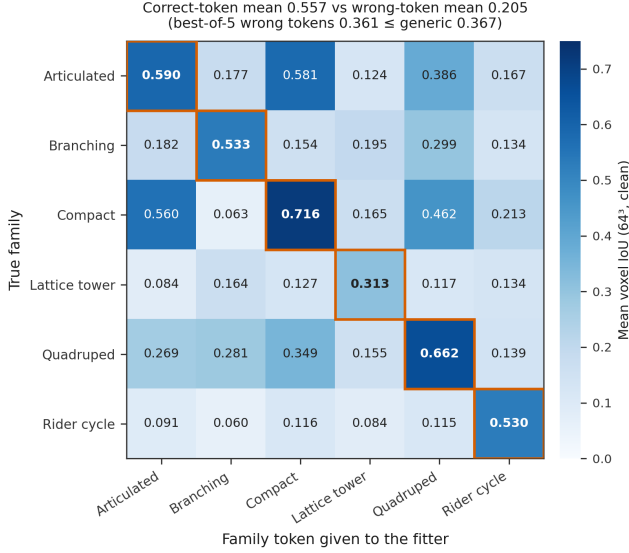


Figure 8: Wrong-family analysis (clean condition, $n=240$, mean voxel IoU). Rows: true family; columns: family token supplied to the fitter. The diagonal is the correct token. Off-diagonal cells fall below the diagonal and, with few exceptions, also below the generic-graph mean of 0.367; the exceptions cluster on the morphologically close vehicle families (compact/articulated) and on the quadruped token applied to vehicle rows.

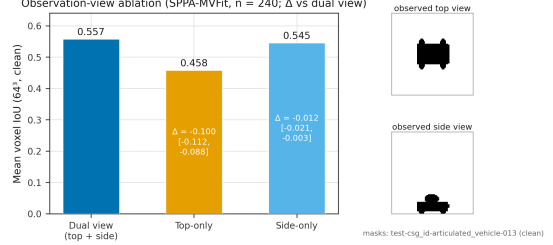


Figure 9: View ablation of SPPA-MVFit (clean condition, $n=240$, correct family token): dual-view, top-only, and side-only mean voxel IoU with 95% bootstrap intervals.

4.5 Adversarial family instances

A structural concern with the primary comparison is that it is nearly guaranteed: the family graph encodes the same part topology that the CSG-ID generators instantiate, so a positive H1 could be decided by design rather than by measurement. The wrong-family arm already shows the prior can be actively harmful; here we probe the complementary edge. A post-hoc exploratory stratum of 120 adversarial actors (20 per family, twelve documented violation types, design frozen before measurement) keeps each family’s semantic class but violates its structural prior: rearward cabs, split cargo beds, six-legged and asymmetric quadrupeds, out-of-order platforms, a 25° leaning lattice tower, sidecar cycles, and cargo over-scaled to 250% of the prior. Under the identical clean-mask protocol, the paired SPPA advantage over Generic-MVFit falls from $+0.209$ on the same actors unviolated to $+0.141$ (CI [$0.125, 0.157$]): violating the prior destroys about one third of the advantage ($\Delta\Delta = -0.068$, CI [$-0.087, -0.048$]) and SPPA-MVFit loses at actor level in 8.3% of cases. Two violation types reach statistical ties with both methods near the floor—the leaning tower ($+0.009$, CI [$-0.008, +0.026$]; a 25° shear is not expressible by axis-aligned part scaling) and the cascade crown ($+0.007$, CI [$-0.015, +0.029$])—while others leave the advantage intact or larger (sidecar $+0.320$). The family prior is therefore neither a tautological win nor a fragile one: its value degrades measurably and predictably with prior violation, and the failure zones are identifiable (Table in Supplementary Section S.9).

4.6 Ablations and sensitivity analyses

View ablation and side-view acquisition. The objective consumes a top and a side silhouette. Refitting with only the top view costs -0.100 IoU (CI [$-0.112, -0.088$]); refitting with only the side view costs only -0.012 (CI [$-0.021, -0.003$]) (Figure 9). The side view carries nearly all of the 3D information for this object envelope, and even the degraded top-only mode (0.458) stays

above dual-view Generic-MVFit (0.367) and the box baselines. A side view is not available from a pure nadir pass: it requires an oblique pass or orbit, or a second platform. The operational pipeline already runs in a nadir footprint-only mode in production (Section 3.2); the top-only arm quantifies exactly that degraded mode for the fitting path.

Oriented bounding box baseline. The primitive baselines are axis-aligned. A rotation-aware oriented bounding box (top-mask minimum-area rectangle with gap-midpoint refinement, z extent from the side mask) scores 0.252, practically identical to the axis-aligned box (0.248; paired difference -0.004 , CI $[-0.005, -0.003]$): the benchmark actors are presented axis-aligned, so orientation freedom adds nothing, and SPPA-MVFit minus OBB is $+0.306$ (CI $[0.297, 0.314]$; table in Supplementary Section S.9). The box family of baselines, oriented or not, does not close the representation gap. The wider representation design space is mapped in Supplementary Section S.1.

Generic-graph design sensitivity. Because the authors designed the generic graph (Section 3.4), three alternative generic graphs were registered in writing before any measurement and evaluated under the same protocol: G2 (box/cylinder chassis), G3 (vertical ellipsoid stack plus legs), and G4 (slot-wise mean of the six family graphs, a fully mechanical rule). All three score *below* the original generic (table in Supplementary Section S.9): against these alternatives the measured SPPA advantage is, if anything, conservative.

Optimizer budget. The 31-candidate budget sits at the knee of the budget curve: 11 candidates give 0.528 at 3.6 ms mean, 21 give 0.547 at 7.6 ms, the adopted 31 give 0.557 at 12.6 ms, and doubling to 61 gives 0.560 at 24.0 ms—a non-significant $+0.002$ (CI $[-0.001, +0.005]$) for twice the compute (full sweep in Supplementary Section S.9). Sweep timings come from a separate post-hoc re-run on the same machine; IoU values are bit-exact with the sealed run, which reports a 9.4 ms median.

Role-aware IoU. Occupancy IoU does not test the role labels themselves. As a first quantitative check, a slot-to-component mapping against the ground-truth generators was frozen in writing before any computation and restricted ex-ante to the CSG-ID stratum (120 actors, 920 matched slot-component pairs). Matched pairs reach 0.545 coverage and a role IoU of 0.319 (Figure 3b), against 0.053 for a random-shuffle control (cyclic shuffle 0.017): a paired role-assignment advantage of $+0.265$ (CI $[0.250, 0.281]$; table in Supplementary Section S.9). This is descriptive evidence on a frozen mapping over one stratum, not a confirmatory endpoint.

Part-query task. To operationalize what role labels are worth, a part-query task was frozen in writing before measurement on the same stratum: given a target role (*cargo, wheels, cab, legs, hitch, platforms*), each method must return the voxels of that part. SPPA-MVFit returns its role-labeled slot; controls are Generic-MVFit with its own slot mapping and two frozen visual-hull heuristics (largest connected component; lower half). Over 887 scorable queries (33 of 920 excluded for sub-voxel ground truth, documented with a sensitivity analysis), SPPA-MVFit reaches query F1 0.434 versus 0.145 (generic), 0.111 and 0.087 (hull heuristics), with normalized part-centroid error 0.055 versus 0.177–0.249; the hull *covers* (recall 0.96) but cannot *select* (precision 0.07). Structural part counting is exact in 70% of actors versus 0% for the hull. The frontier is honest: the largest-component heuristic *wins* on fused branching crowns, the rider fork scores zero, and lattice platform counts fail where graph arity and ground truth disagree. Role value concentrates on small, numerous parts (table in Supplementary Section S.9).

4.7 External sanity check on real meshes

As a post-hoc exploratory case study outside the design distribution, the same methods were run on $n=52$ real third-party meshes (Objaverse LVIS subset and ModelNet40), mapped to the six

families through a class-to-family mapping frozen in writing before any evaluation and passed through visual quality control (Table 5; Figures 10 and 11). Mean fit time remains millisecond-scale (9.8 ms per case, CPU). This is a sanity check, not a benchmark: the sample is small, the class mapping is approximate, and the evaluation reuses the synthetic-grid protocol on real geometry.

Table 5: External sanity check on 52 real meshes (voxel IoU at 64^3 , clean condition; family headers abbreviated: compact/articulated vehicle, branching vertical, lattice tower, rider cycle). Mapping class-to-family frozen before evaluation; per-family n in the table footer. The preregistered $+0.030$ margin does not replicate here (paired SPPA–Generic $+0.043$, CI $[-0.007, +0.094]$).

Method	compact	articulated	quadruped	branching	lattice	rider	All
SPPA-MVFit	0.632	0.633	0.423	0.215	0.157	0.361	0.413
Generic-MVFit	0.508	0.319	0.393	0.382	0.288	0.288	0.370
SPPA text-only	0.449	0.188	0.242	0.058	0.101	0.172	0.213
AABB	0.577	0.593	0.386	0.244	0.368	0.259	0.411
Ellipsoid	0.613	0.527	0.496	0.369	0.470	0.398	0.485
Capsule	0.654	0.605	0.477	0.365	0.472	0.342	0.492
Billboard	0.164	0.070	0.256	0.111	0.077	0.333	0.172
Visual hull	0.809	0.783	0.715	0.462	0.581	0.530	0.656

n per family: compact=10, articulated=8, quadruped=10, branching=8, lattice=8, rider=8 (total 52).
Overall mean [CI95]: SPPA-MVFit 0.413 [0.359, 0.467]; Generic 0.370; Visual hull 0.656.
Paired SPPA–Generic: $+0.043$ $[-0.007, +0.094]$.

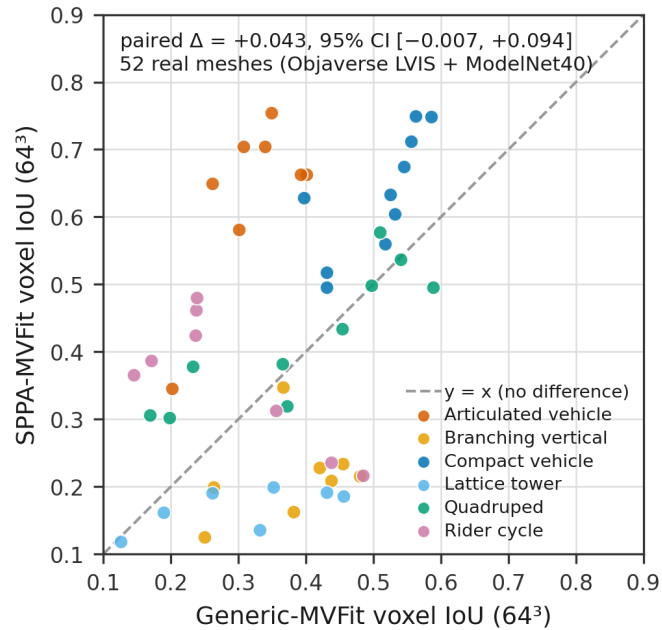


Figure 10: External sanity check: per-case SPPA-MVFit versus Generic-MVFit voxel IoU on the 52 real meshes, colored by mapped family. Points near the diagonal dominate; the paired difference ($+0.043$, CI $[-0.007, +0.094]$) includes zero.

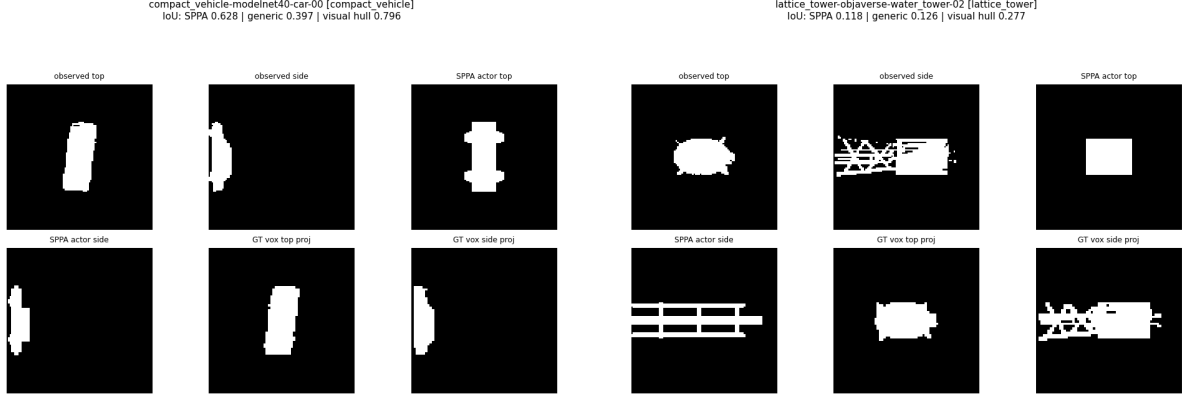


Figure 11: External sanity check, qualitative. Left: a ModelNet40 car where the family graph helps (SPPA 0.628 vs generic 0.397). Right: an Objaverse water tower whose horizontal tank mismatches the vertical tower template (SPPA 0.118 vs generic 0.126)—the template–instance mismatch failure mode. Panels show observed masks, SPPA actor projections, and ground-truth voxel projections.

Three readings follow. First, the preregistered margin does not replicate on external meshes at this sample size: SPPA-MVFit scores 0.413 (CI [0.359, 0.467]) against Generic-MVFit 0.370 (CI [0.335, 0.405]), a paired +0.043 whose interval $[-0.007, +0.094]$ includes zero. Second, the nonsemantic visual hull dominates raw occupancy with clean real silhouettes (0.656, CI [0.606, 0.704]), and the trivial geometric primitives tell the same story: capsule (0.492), ellipsoid (0.485), and even the axis-aligned box (0.411) match or exceed SPPA-MVFit (0.413) here—outside the design distribution the family graph buys no volumetric advantage over shape-free baselines, and the value of SPPA is the semantic contract: role labels, bounded updates, and a telemetry-scale payload. Third, the failure modes are informative and consistent with Section 4.4: template–instance mismatch (horizontal water towers fitted by a vertical tower graph), open geometry (bicycle wheel spokes), and one LVIS mislabel retained as an out-of-distribution case (a giraffe labeled *horse*). When the prior does not match, it hurts; the contract’s role is to detect that situation and degrade safely. Under mild morphology corruption the pattern persists (SPPA 0.380, Generic 0.357).

4.8 External neural generators: an input-modality mismatch demonstration

A secondary analysis, specified in writing before any measurement, compares SPPA-MVFit with two measured external neural image-to-3D generators, TripoSR [51] and Hunyuan3D-2mini-turbo [63], on sixty of the held-out cases (Figure 12; full table and protocol in Supplementary Section S.2). Each generator received two inputs: a clean-crop shaded oblique render, its best condition, and the raw 96×96 top telemetry mask, the closest visual analog of the channel SPPA consumes. On these cases SPPA-MVFit reaches 0.561 mean voxel IoU at 9.2 ms CPU with a 1.45 kB update descriptor, against 0.128–0.231 for the measured neural outputs, whose meshes carry 28k–3.1M triangles and MB-scale payloads on an RTX 5090. Two observations sharpen the boundary. First, both generators score *higher* from the harsh top mask than from their generous clean-crop render: single-view image-to-3D models hallucinate occluded structure, whereas a calibrated footprint constrains occupancy directly—telemetry-shaped evidence is not a degraded image but a different measurement channel. Second, the wave measures one operating point per system under deliberately asymmetric input channels: Hunyuan3D-2mini-turbo is a distilled mini/turbo checkpoint, not the flagship model of its family, and the comparison is not an image-to-3D leaderboard.

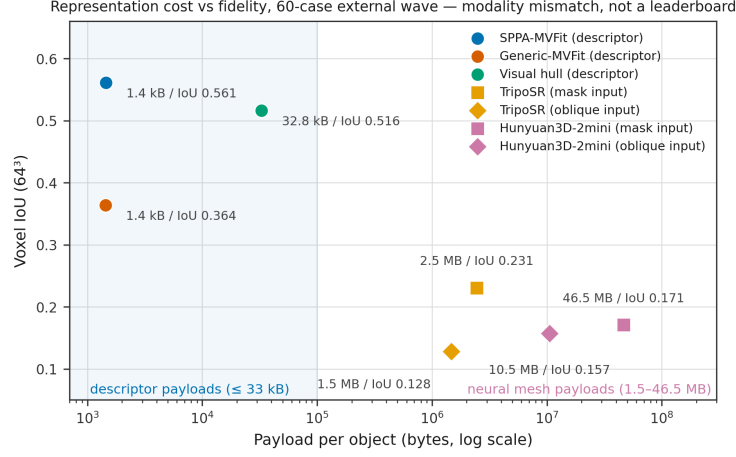


Figure 12: Operating-point comparison on sixty held-out cases: mean voxel IoU against per-object payload in bytes (log scale) for SPPA-MVFit, Generic-MVFit, visual hull, and the measured neural generators under both input conditions. SPPA rows are CPU-only and their payload is the update descriptor, not a mesh; the juxtaposition is an input-modality mismatch shown for operating-point context, not a leaderboard.

4.9 Real-image probes and deployment runtime

Four real-image probes exercise the dual-input contract of Section 3.2 (Figure 13): a YOLOE-26s open-vocabulary detector supplies approximate semantic evidence on the image path, while the tag path uses reviewed text only. Detector evidence is not treated as ground truth: weak labels are refined only to conservative families (YOLOE’s `vehicle` on the tractor-trailer crop becomes `articulated_vehicle`, never the reviewed `tractor_trailer` class), implausible footprint aspect ratios are clamped by family priors, and unsafe image-derived pose is withheld from the runtime descriptor. A 20-label verifier-gated open-label probe accepts 16/20 labels into reviewed families with 8/20 verified recipes (Supplementary Section S.3), and a role-preservation diagnostic keeps cab and wheel priors fixed while cargo length changes (Supplementary Section S.5). Per-probe evidence, replay audits, and input-mode ablations are in Supplementary Section S.4; no 3D ground truth exists for these images, so no IoU is reported for them.

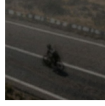
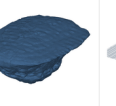
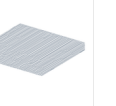
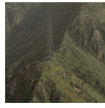
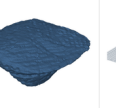
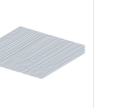
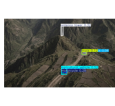
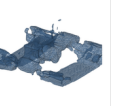
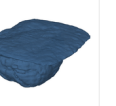
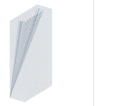
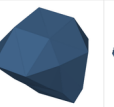
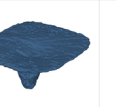
	Input evidence real crop, not GT	Detector probe YOLOE image evidence	SPPA complete proxy	Shap-E text-only workflow	Point-E text-only workflow	TripoSr image/crop workflow	Hunyuan3D image/crop workflow
biker		 YOLOE mask: person+motorcycle	 tag: cyclistbiker obs fused soft scale fusion gen 3.3 ms 1,056 tris	 gen 4.44 s 16,928 tris VRAM 6.0 GB	 gen 11.77 s 638 tris VRAM 3.6 GB	 gen 998.0 ms 22,436 tris VRAM 2.0 GB	 gen 2.75 s 592,008 tris VRAM 5.2 GB
tower		 YOLOE mask: electric pylon	 tag: towerpylon obs fused height-only obs gen 2.6 ms 420 tris	 gen 1.61 s 7,964 tris VRAM 6.0 GB	 gen 6.88 s 144 tris VRAM 3.6 GB	 gen 1.03 s 24,228 tris VRAM 2.0 GB	 gen 2.39 s 592,204 tris VRAM 5.2 GB
tractor		 YOLOE mask: agricultural vehicle	 tag: tractorfarm obs fused aspect fusion gen 3.5 ms 1,095 tris	 gen 2.71 s 119,558 tris VRAM 6.0 GB	 gen 6.67 s 1,594 tris VRAM 3.6 GB	 gen 1.35 s 36,448 tris VRAM 2.0 GB	 gen 3.09 s 870,568 tris VRAM 5.2 GB
tractor+trailer		 YOLOE mask: vehicle	 tag: tractor_trailer obs fused soft scale fusion gen 3.2 ms 2,012 tris	 gen 2.30 s 52,168 tris VRAM 6.0 GB	 gen 6.88 s 40 tris VRAM 3.6 GB	 gen 1.20 s 27,944 tris VRAM 2.0 GB	 gen 2.48 s 725,376 tris VRAM 5.2 GB

Figure 13: Real-image probes for biker, electrical tower, tractor, and tractor-with-trailer (rows). Columns: input crop, YOLOE detector evidence, the emitted SPPA proxy, and four reproduced text/image-to-3D baselines (Shap-E, Point-E, TripoSr, Hunyuan3D) annotated with generation time, triangle count, and VRAM. Detector output is evidence, not ground truth; the neural columns receive a single image or crop, an input-modality mismatch shown for operating-point context, not a ranking.

The deployment layer is the descriptor/update contract consumed by the optional Unreal Engine 5.7 backend. Table 6 condenses the runtime evidence and Figure 14 the packaged scaling behavior; the full Unreal/HISM matrix with run identifiers is Supplementary Section S.8. Packaged replay at 100 synthetic objects passes a 33.3 ms desktop frame budget but misses 60 FPS; dense shape updates remain the limiting factor even under HISM batching, and curated assets remain preferable when a matched asset exists (shape-stream P95 6.789 ms vs. 22.445 ms on the two-label cow/tower subset).

Table 6: Condensed deployment runtime evidence (run identifiers omitted here; full matrix in Supplementary Section S.8). Python contract timings are not Unreal frame times, and packaged runs are short synthetic replays, not live flight telemetry.

Evidence	Summary value	Boundary
Descriptor build (81,568-row replay)	P50/P95 391.8/515.0 μ s	Python contract cost, not frame time
Update-packet build	P50/P95 16.8/75.3 μ s	Python contract cost
Scheduler decision	P50/P95 9.2/14.2 μ s	Python contract cost
Editor-Cmd and packaged recorded replay	27/27 payload applications accepted	recorded streams, no live network
Packaged 100-object replay	pose/shape frame P95 17.814/19.256 ms	desktop feasibility; misses 60 FPS P95
Packaged HISM partial update, 500 objects	pose/shape frame P95 16.528/29.787 ms	10% changed-track synthetic schedule

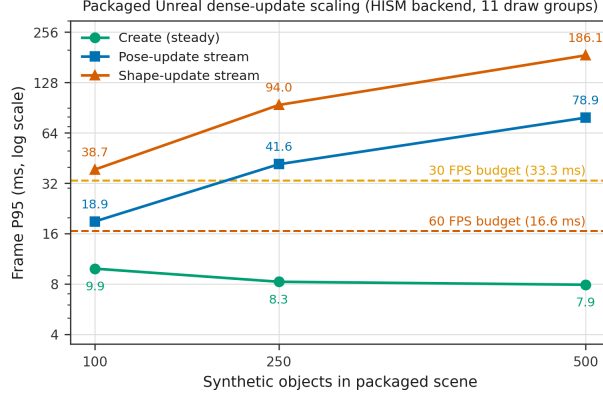


Figure 14: Packaged-runtime scaling (HISM backend, dense-update streams): frame P95 versus object count for the create, pose-update, and shape-update streams. The 33.3 ms line marks a desktop 30 FPS budget; dense shape updates exceed it already at 100 objects and dense pose updates between 100 and 250, while the sparse 10% changed-track schedule of Table 6 stays below it at 500 objects.

4.10 A real detector stream case study

The motivating scenario is a real detector running on a live georeferenced twin. As a post-hoc exploratory case study, a recorded 1,394-frame flight stream over the Ejea twin was re-analyzed end to end: a custom YOLO detector with real confidence noise and real confusions (cows hallucinated on tower textures), MAVLink telemetry, and exact simulator ground truth for the static tower anchors. Detections with locked telemetry yield 1,902 cases (308 tower, 848 cow, 746 biker detections). Every method receives the identical observation—an oriented ground footprint geo-projected from the detection box plus a monocular height estimate—so the comparison carries no input advantage, and wrong detector tokens are kept as a natural condition.

Three findings bound the operating point honestly (Table 7; Figure 15). First, localization is observation-bound, not fitter-bound: all methods sit at ≈ 33 m median 3D error, exactly on the identity line of footprint-center error versus anchor error, because the low-altitude stream (AGL 10–23 m) compresses monocular range; footprint IoU against a 5×5 m base is degenerate for every method. Second, SPPA-MVFit does *not* win the 2D reprojection metric on real evidence (0.298 median versus 0.42–0.45 for box proxies): the family prior buys 3D part structure at the cost of image overlap when the monocular height estimate conflicts with the prior bounds (a 13.75 m tower column versus a ≈ 5 m observed extent). The stream carries no 3D ground truth, so the quantity the sealed benchmark measures—3D occupancy—cannot be scored here; what the stream adds is precisely the evidence-versus-prior tension that a ground-truth benchmark cannot see. Third, the tension is quantifiable: on the 138 GT-matched cases where the real detector emitted a wrong family token, refitting with the correct token collapses the 2D fit from 0.381 to 0.025—the correct prior cannot explain the misleading evidence that triggered the wrong label (token-arm table in Supplementary Section S.9). Fit latency (11.8 ms median) stays far below the 2–15 Hz stream rate, so the operating point is computationally viable even in this degraded mode.

We read this study as the field counterpart of the boundary map: in production the system should run the footprint-only mode declared in Section 3.2 (and quantified by the top-only arm), reserve dual-view fitting for oblique passes, and treat detector-token confidence as a routing signal rather than a given.

Table 7: Real detector stream case study (exploratory post-hoc, not sealed): 1,902 detections from a 1,394-frame flight over the georeferenced twin with a real custom detector, real telemetry, and exact tower-anchor ground truth. Localization and footprint columns: GT-matched subset ($n=217$; wrong detector tokens kept as a natural condition); reprojection and latency: all cases. All methods are observation-bound at ≈ 33 m; SPPA-MVFit trades 2D image overlap for 3D part structure (the stream has no 3D ground truth).

Method	Loc. err. 3D med. (m) [†]	Footprint IoU [†]	2D reproj. IoU med. [P25, P75]	Latency (ms)
SPPA-MVFit (this work)	32.8	0.000	0.298 [0.218, 0.392]	11.8
Generic-MVFit	32.8	0.000	0.423 [0.394, 0.448]	13.4
OBB	32.7	0.000	0.446 [0.398, 0.492]	0.22
AABB	32.7	0.000	0.330 [0.299, 0.357]	0.25
Visual hull (non-semantic)	32.7	0.000	0.446 [0.398, 0.493]	0.50
Capsule	32.7	0.000	0.445 [0.422, 0.466]	1.31

E7 Real Stream Wave — real UAV stream, real detector, exploratory post-hoc (not sealed)

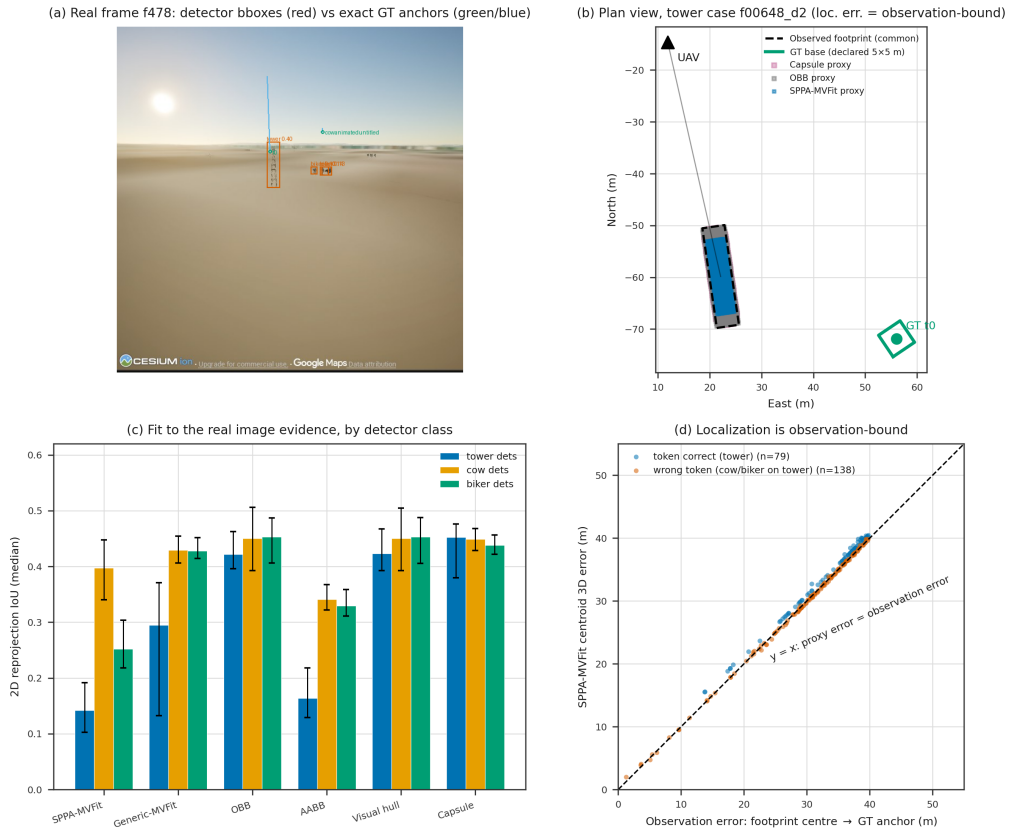


Figure 15: Real stream case study. (a) Frame 478 with detector boxes and exact ground-truth anchors. (b) Plan view of a tower case: the observed footprint (dashed) and all fitted proxies sit ≈ 35 m from the exact anchor—observation-bound. (c) 2D reprojection IoU by detector class. (d) Fitted-centroid error versus observation error: the identity line holds for all methods.

5 Discussion

The confirmatory result answers the preregistered question: under identical calibrated silhouettes and optimizer budget, a frozen family part graph improves lightweight occupancy over a generic eight-part graph by a large and stable margin. The decomposition of Section 4.3 shows that the knowledge representation, not a clever optimizer, is the active ingredient; secondary gains over text-only SPPA show that silhouette evidence helps when the same family graph is held fixed.

The boundary analyses are as informative as the headline. A wrong family token is worse than no prior at all, which turns fallback routing from an engineering convenience into a measured requirement. The adversarial stratum shows the advantage is killable rather than tautological: violating the structural prior while preserving the semantic class destroys a third of the effect and produces identifiable tie zones where no axis-aligned part graph can win. The side view carries nearly all of the 3D information, yet it is unavailable from a pure nadir pass: acquiring it requires an oblique pass, an orbit, or a second platform, and otherwise the system degrades to the footprint-only mode that already runs in production—quantified by the top-only arm, which still beats the generic dual-view fit. On external real meshes the margin does not transfer and shape-free baselines match or exceed SPPA on raw occupancy; what remains differentiating outside the design distribution is the contract—role labels, bounded updates, and a telemetry-scale payload—not volumetric leadership. Visual hull can approach SPPA-MVFit IoU because it is free to materialize denser occupancy without an eight-part role-labeled actor contract; the part-query task quantifies what that contract buys, since the hull covers but cannot select (query F1 0.434 versus 0.111). Finally, the real detector stream relocates the bottleneck entirely: under monocular low-altitude observation every method is observation-bound, and the family prior trades image overlap for 3D structure—so deployment should route between footprint-only and dual-view modes rather than treat fitting as always-on.

Several threats to validity remain binding. The primary comparison is structurally favored by design: the family graphs encode the same part topology that the CSG-ID generators instantiate, so preregistration quantifies the effect but cannot make it surprising—the load-bearing evidence that the advantage is real, killable, and bounded is the boundary map (wrong-family arm, adversarial stratum, external meshes, real stream), not H1 alone. The held-out set is synthetic and developer-held-out; positive OOD-stratum differences do not license claims about real UAV masks, detectors, or metric flight geometry. Family tokens are supplied, so the experiment is family-conditioned rather than open-set. Silhouettes are clean orthographic projections plus prespecified corruptions, not real detector masks. Timing is local CPU wall-clock for the fitter, not flight-laptop Unreal frame time, and the packaged replay evidence consists of short synthetic streams, not live telemetry or VR readiness. The real-stream study is a single recorded flight with one custom detector, monocular low-altitude views, no 3D ground truth, and 2D reprojection metrics only; its localization errors reflect the observation model, not the fitter, and its tower-anchor subset is small. The same team authored the family graphs and the generic competitor; the sensitivity variants of Section 4.6 bound, but do not eliminate, the resulting baseline-design risk. Within these boundaries the supported claim is narrow: on the held-out synthetic test, family conditioning improves occupancy beyond the prespecified margin; the shared generator supports text-only and silhouette-fitted modes; and SPPA descriptors can be consumed by an optional Unreal backend that coexists with curated assets. Real-UAV reconstruction accuracy, universal open-set geometry, operator benefit, and neural image-to-3D ranking remain outside the claim.

Three directions are left to future work. First, the frozen coordinate-descent fitter could be replaced by population-based or differentiable-rendering fitting, and the iterative fit could be amortized by a small learned mask-to- θ predictor, keeping the representation and contract while moving per-case cost toward microseconds; the primitive vocabulary could likewise be extended toward superquadrics. Second, the external gap measured here calls for public real-data tests with metric ground truth and detector-derived silhouettes rather than assumed calibration. Third, any operator-facing benefit must be evaluated with human-factors methods such as SAGAT and NASA-TLX [9, 10, 15]; the present paper contains no pilot study.

6 Conclusion

Family-conditioned SPPA-MVFit shows that a reviewed semantic part graph is a measurable occupancy prior for lightweight multiview proxy fitting. Under a preregistered equal-budget comparison, SPPA-MVFit improves clean 3D occupancy over Generic-MVFit by 0.190 absolute IoU (95%

CI [0.181, 0.199]) on 240 held-out synthetic actors, with positive strata and millisecond-scale local cost. Most of the advantage is the family-graph representation rather than the fitter; a wrong family token is worse than no family prior at all; and the margin does not transfer unchanged to real meshes whose instances mismatch the templates. Adversarial instances that violate the structural prior destroy a third of the advantage and expose identifiable tie zones, while a real detector stream with exact anchor ground truth relocates the field bottleneck in the monocular observation itself. The durable contribution is the contract—family-conditioned roles, bounded updates, and a telemetry-scale payload—whose part-query value is measured (query F1 0.434 versus 0.111 for occupancy-only controls), together with a quantified occupancy envelope that holds inside the design distribution. The next steps are external public data with metric ground truth, detector-derived silhouettes, and, only if claimed, operator studies.

Data Availability

The SPPA-MVFit reproducibility package, protocol amendments, NIST-bound seed manifest hashes, and confirmatory summaries are included under `reproducibility/sppa_mvfit/`. Large Unreal artifacts and historical engineering logs remain supporting material and are not required to recompute the primary H1 tables.

Code Availability

The fitting, evaluation, and table-export code used to produce the reported results is part of the reproducibility package above and will be released in a public repository upon publication.

References

- [1] Iro Armeni, Zhi-Yang He, JunYoung Gwak, Amir R. Zamir, Martin Fischer, Jitendra Malik, and Silvio Savarese. 3d scene graph: A structure for unified semantics, 3d space, and camera. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5664–5673, 2019.
- [2] Michael Batty. Digital twins. *Environment and Planning B: Urban Analytics and City Science*, 45(5):817–820, 2018.
- [3] Irving Biederman. Recognition-by-components: A theory of human image understanding. *Psychological Review*, 94(2):115–147, 1987.
- [4] Volker Blanz and Thomas Vetter. A morphable model for the synthesis of 3D faces. In *Proceedings of ACM SIGGRAPH*, pages 187–194, 1999.
- [5] Mark Boss, Zixuan Huang, Aaryaman Vasishta, and Varun Jampani. Sf3d: Stable fast 3d mesh reconstruction with uv-unwrapping and illumination disentanglement. *arXiv preprint arXiv:2408.00653*, 2024.
- [6] Garrick Brazil, Abhinav Kumar, Julian Straub, Nikhila Ravi, Justin Johnson, and Georgia Gkioxari. Omni3d: A large benchmark and model for 3d object detection in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- [7] James H. Clark. Hierarchical geometric models for visible surface algorithms. *Communications of the ACM*, 19(10):547–554, 1976.
- [8] Xavier Decoret, Fredo Durand, Francois X. Sillion, and Julie Dorsey. Billboard clouds for extreme model simplification. *ACM Transactions on Graphics*, 22(3):689–696, 2003.
- [9] Mica R. Endsley. Situation awareness global assessment technique (sagat). In *Proceedings of the IEEE 1988 National Aerospace and Electronics Conference*, 1988.
- [10] Mica R. Endsley. Toward a theory of situation awareness in dynamic systems. *Human Factors*, 37(1):32–64, 1995.
- [11] Epic Games. Nanite virtualized geometry overview, 2026. Rolling Unreal Engine documentation; accessed 2026-07-15.
- [12] Elisabetta Fedele, Francis Engelmann, Ian Huang, Or Litany, Marc Pollefeys, and Leonidas J. Guibas. Space-Control: Introducing test-time spatial control to 3D generative modeling. *arXiv preprint arXiv:2512.05343*, 2025.

- [13] Elisabetta Fedele, Boyang Sun, Leonidas Guibas, Marc Pollefeys, and Francis Engelmann. Superdec: 3d scene decomposition with superquadric primitives. *arXiv preprint arXiv:2504.00992*, 2025.
- [14] Aditya Ganeshan, Matheus Gadelha, Thibault Groueix, Zhiqin Chen, Siddhartha Chaudhuri, Vladimir Kim, Wang Yifan, and Daniel Ritchie. Residual primitive fitting of 3d shapes with superfrusta. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7404–7413, 2026.
- [15] Sandra G. Hart and Lowell E. Staveland. Development of nasa-tlx (task load index): Results of empirical and theoretical research. In Peter A. Hancock and Najmedin Meshkati, editors, *Human Mental Workload*, pages 139–183. North-Holland, Amsterdam, 1988.
- [16] Seamie Hayes, Alexandre Boulch, Andrei Bursuc, Reenu Mohandas, Ganesh Sistu, Tim Brophy, and Ciarán Eising. SuperQuadricOcc: Real-time self-supervised semantic occupancy estimation with superquadric volume rendering. *arXiv preprint arXiv:2511.17361*, 2025.
- [17] Armin Hornung, Kai M. Wurm, Maren Bennewitz, Cyrill Stachniss, and Wolfram Burgard. Octomap: An efficient probabilistic 3d mapping framework based on octrees. *Autonomous Robots*, 34(3):189–206, 2013.
- [18] Xuyi Hu, Jin Lyu, Jiuming Liu, Yebin Liu, Silvia Zuffi, Liang An, and Stefan Goetz. Sam 3d animal: Promptable animal 3d reconstruction from images in the wild. *arXiv preprint arXiv:2605.07604*, 2026.
- [19] Nathan Hughes, Yun Chang, and Luca Carlone. Hydra: A real-time spatial perception system for 3D scene graph construction and optimization. In *Robotics: Science and Systems*, 2022.
- [20] Hyper3D. Hyper3d rodin gen-2.5, 2026.
- [21] R. Kenny Jones, Theresa Barton, Xianghao Xu, Kai Wang, Ellen Jiang, Paul Guerrero, Niloy J. Mitra, and Daniel Ritchie. Shapeassembly: Learning to generate programs for 3d shape structure synthesis. *ACM Transactions on Graphics*, 39(6), 2020.
- [22] R. Kenny Jones, Paul Guerrero, Niloy J. Mitra, and Daniel Ritchie. Shapecoder: Discovering abstractions for visual programs from unstructured primitives. *ACM Transactions on Graphics*, 42(4), 2023.
- [23] Heewoo Jun and Alex Nichol. Shap-e: Generating conditional 3d implicit functions. *arXiv preprint arXiv:2305.02463*, 2023.
- [24] Mohammad Sadil Khan, Sankalp Sinha, Talha Uddin Sheikh, Didier Stricker, Sk Aziz Ali, and Muhammad Zeshan Afzal. Text2cad: Generating sequential cad designs from beginner-to-expert level text prompts. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- [25] Aldo Laurentini. The visual hull concept for silhouette-based image understanding. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 16(2):150–162, 1994.
- [26] Dong-Yang Li, Wang Zhao, Yuxin Chen, Wenbo Hu, Meng-Hao Guo, Fang-Lue Zhang, Ying Shan, and Shi-Min Hu. Pixel3d: Pixel-aligned 3d generation from images. *arXiv preprint arXiv:2605.10922*, 2026.
- [27] Yangguang Li, Zi-Xin Zou, Zexiang Liu, Dehu Wang, Yuan Liang, Zhipeng Yu, Xingchao Liu, Yuan-Chen Guo, Ding Liang, Wanli Ouyang, and Yan-Pei Cao. Triposg: High-fidelity 3d shape synthesis using large-scale rectified flow models. *arXiv preprint arXiv:2502.06608*, 2025.
- [28] Zizhang Li, Dor Litvak, Ruining Li, Yunzhi Zhang, Tomas Jakab, Christian Rupprecht, Shangzhe Wu, Andrea Vedaldi, and Jiajun Wu. Learning the 3d fauna of the web. *arXiv preprint arXiv:2401.02400*, 2024.
- [29] Chen-Hsuan Lin, Jun Gao, Luming Tang, Towaki Takikawa, Xiaohui Zeng, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, and Tsung-Yi Lin. Magic3d: High-resolution text-to-3d content creation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- [30] Yuchen Lin, Chenguo Lin, Panwang Pan, Honglei Yan, Yiqiang Feng, Yadong Mu, and Katerina Fragkiadaki. Partcrafter: Structured 3d mesh generation via compositional latent diffusion transformers. *arXiv preprint arXiv:2506.05573*, 2025.
- [31] Weihang Liu, Yuhui Zhong, Yuke Li, Xi Chen, Jiadi Cui, Honglong Zhang, Lan Xu, Xin Lou, Yujiao Shi, Jingyi Yu, and Yingliang Zhang. CityGo: Lightweight urban modeling and rendering with proxy buildings and residual gaussians. *arXiv preprint arXiv:2505.21041*, 2025.
- [32] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: A skinned multi-person linear model. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia)*, 34(6):248:1–248:16, 2015.
- [33] David Marr and H. Keith Nishihara. Representation and recognition of the spatial organization of three-dimensional shapes. *Proceedings of the Royal Society of London. Series B, Biological Sciences*, 200(1140):269–294, 1978.
- [34] Xiaoxu Meng, Zhongmin Chen, Bo Yang, Weikai Chen, Weixiao Liu, and Lin Gao. Dualprim: Compact 3d reconstruction with positive and negative primitives. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 29081–29091, 2026.
- [35] Kaichun Mo, Shilin Zhu, Angel X. Chang, Li Yi, Subarna Tripathi, Leonidas J. Guibas, and Hao Su. PartNet: A large-scale benchmark for fine-grained and hierarchical part-level 3D object understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 909–918, 2019.

- [36] Pascal Müller, Peter Wonka, Simon Haegler, Andreas Ulmer, and Luc Van Gool. Procedural modeling of buildings. *ACM Transactions on Graphics*, 25(3):614–623, 2006.
- [37] Alex Nichol, Heewoo Jun, Prafulla Dhariwal, Pamela Mishkin, and Mark Chen. Point-e: A system for generating 3d point clouds from complex prompts. *arXiv preprint arXiv:2212.08751*, 2022.
- [38] Helen Oleynikova, Zachary Taylor, Marius Fehr, Juan Nieto, and Roland Siegwart. Voxblox: Incremental 3d euclidean signed distance fields for on-board mav planning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 1366–1373, 2017.
- [39] Open Geospatial Consortium. Ogc 3d tiles 1.1 community standard, 2023.
- [40] Despoina Paschalidou, Ali Osman Ulusoy, and Andreas Geiger. Superquadrics revisited: Learning 3d shape parsing beyond cuboids. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.
- [41] Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. Dreamfusion: Text-to-3d using 2d diffusion. *arXiv preprint arXiv:2209.14988*, 2022.
- [42] Antoni Rosinol, Arjun Gupta, Marcus Abate, Jingnan Shi, and Luca Carlone. 3d dynamic scene graphs: Actionable spatial perception with places, objects, and humans. In *Robotics: Science and Systems*, 2020.
- [43] Danila Rukhovich, Elona Dupont, Dimitrios Mallis, Kseniya Cherenkova, Anis Kacem, and Djamila Aouada. Cad-recode: Reverse engineering cad code from point clouds. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025.
- [44] SAM 3D Team, Xingyu Chen, Fu-Jen Chu, Pierre Gleize, Kevin J. Liang, Alexander Sax, Hao Tang, Weiyao Wang, Michelle Guo, and Thibaut Hardin. SAM 3D: 3Dfy anything in images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. to appear; identifier pending verification.
- [45] Gopal Sharma, Rishabh Goyal, Difan Liu, Evangelos Kalogerakis, and Subhansu Maji. Csgnet: Neural shape parser for constructive solid geometry. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018.
- [46] Franc Solina and Ruzena Bajcsy. Recovery of parametric models from range images: The case for superquadrics with global deformations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 12(2):131–147, 1990.
- [47] Stability AI. Introducing stable fast 3d: Rapid 3d asset generation from single images, 2024.
- [48] Stability AI. Spar3d: Stable point-aware reconstruction of 3d objects from single images, 2025.
- [49] Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. Lgm: Large multi-view gaussian model for high-resolution 3d content creation. *arXiv preprint arXiv:2402.05054*, 2024.
- [50] Team Hunyuan3D. Hunyuan3d 2.1: From images to high-fidelity 3d assets with production-ready pbr material. *arXiv preprint arXiv:2506.15442*, 2025.
- [51] Dmitry Tochilkin, David Pankratz, Zexiang Liu, Zixuan Huang, Adam Letts, Yangguang Li, Ding Liang, Christian Laforte, Varun Jampani, and Yan-Pei Cao. Tripotr: Fast 3d object reconstruction from a single image. *arXiv preprint arXiv:2403.02151*, 2024.
- [52] Shubham Tulsiani, Hao Su, Leonidas J. Guibas, Alexei A. Efros, and Jitendra Malik. Learning shape abstractions by assembling volumetric primitives. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2635–2643, 2017.
- [53] Ao Wang, Lihao Liu, Hui Chen, Zijia Lin, Jungong Han, and Guiguang Ding. Yoloe: Real-time seeing anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 24591–24602, 2025.
- [54] Zhengyi Wang, Yikai Wang, Yifei Chen, Chendong Xiang, Shuo Chen, Dajiang Yu, Chongxuan Li, Hang Su, and Jun Zhu. Crm: Single image to 3d textured mesh with convolutional reconstruction model. *arXiv preprint arXiv:2403.05034*, 2024.
- [55] Kailu Wu, Fangfu Liu, Zhihan Cai, Runjie Yan, Hanyang Wang, Yating Hu, Yueqi Duan, and Kaisheng Ma. Unique3d: High-quality and efficient 3d mesh generation from a single image. *arXiv preprint arXiv:2405.20343*, 2024.
- [56] Shuang Wu, Youtian Lin, Feihu Zhang, Yifei Zeng, Yikang Yang, Yajie Bao, Jiachen Qian, Siyu Zhu, Philip Torr, Xun Cao, and Yao Yao. Direct3d-s2: Gigascale 3d generation made easy with spatial sparse attention. *arXiv preprint arXiv:2505.17412*, 2025.
- [57] Shun-Cheng Wu, Johanna Wald, Keisuke Tateno, Nassir Navab, and Federico Tombari. Scenegrphfusion: Incremental 3d scene graph prediction from rgb-d sequences. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021.
- [58] Jianfeng Xiang, Xiaoxue Chen, Sicheng Xu, Ruicheng Wang, Zelong Lv, Yu Deng, Hongyuan Zhu, Yue Dong, Hao Zhao, Nicholas Jing Yuan, and Jiaolong Yang. Native and compact structured latents for 3d generation. *arXiv preprint arXiv:2512.14692*, 2025.

- [59] Jianfeng Xiang, Zelong Lv, Sicheng Xu, Yu Deng, Ruicheng Wang, Bowen Zhang, Dong Chen, Xin Tong, and Jiaolong Yang. Structured 3d latents for scalable and versatile 3d generation. *arXiv preprint arXiv:2412.01506*, 2024.
- [60] Jiale Xu, Weihao Cheng, Yiming Gao, Xintao Wang, Shenghua Gao, and Ying Shan. Instantmesh: Efficient 3d mesh generation from a single image with sparse-view large reconstruction models. *arXiv preprint arXiv:2404.07191*, 2024.
- [61] Jin Yao, Hao Gu, Xuweiyi Chen, Jiayun Wang, and Zezhou Cheng. Open vocabulary monocular 3d object detection. *arXiv preprint arXiv:2411.16833*, 2024.
- [62] Hang Yu, Yufei Xu, Jing Zhang, Wei Zhao, Ziyu Guan, and Dacheng Tao. Ap-10k: A benchmark for animal pose estimation in the wild. In *Advances in Neural Information Processing Systems*, 2021.
- [63] Zibo Zhao, Zeqiang Lai, Qingxiang Lin, Yunfei Zhao, Haolin Liu, et al. Hunyuan3d 2.0: Scaling diffusion models for high resolution textured 3d assets generation. *arXiv preprint arXiv:2501.12202*, 2025.
- [64] Chuhan Zou, Ersin Yumer, Jimei Yang, Duygu Ceylan, and Derek Hoiem. 3d-prnn: Generating shape primitives with recurrent neural networks. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 900–909, 2017.
- [65] Silvia Zuffi, Angjoo Kanazawa, David W. Jacobs, and Michael J. Black. 3D menagerie: Modeling the 3D shape and pose of animals. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6365–6373, 2017.